

AMS Onboarding Design Review

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A dark blue diagonal gradient bar that starts from the bottom left corner and extends towards the top right corner, covering the lower half of the slide.

Design Analysis – Oscillator

- Using current starved inverters for tunable frequency and the ability to achieve lower frequencies with less stages than standard inverters (I chose 3 stages)
- Need bias voltages such that the starve current through the PFETs and NFETs are equal
 - Need gate voltage to produce I_{BIAS} thru an NFET + need the current copied for PFET mirror → can use an NFET current mirror
 - Use output current from NFET current mirror as input to PFET current mirror
 - Don't actually care about the output of the PFET current mirror, just need the gate voltage that allows the current to match the input, so the second PFET in the mirror can be eliminated

Design Analysis – Calculating I_{starve}

- Target frequency is 500 MHz $\rightarrow t_d$ per inverter stage is 333 ps
- $I_{\text{starve}} = C_{\text{LOAD}} V_{\text{DD}} / 2t_d$
- Between stages A and B: $C_{\text{LOAD}} \approx C_{\text{OUT, A}} + C_{\text{IN, B}}$ (ignoring parasitic from wire for approximation)
- $C_{\text{IN, B}} = (C_{\text{OX}} * W_n * L_n) + (C_{\text{OX}} * W_p * L_p)$
 - Set W and L for switching FETs to default values in Cadence (550n x 350n for PFET, 420n x 150n for NFET)
 - $C_{\text{OX}} \sim 8.3 \text{ fF/um}$ for Sky130
 - $C_{\text{IN, B}} = 2.12 \text{ fF}$
- $C_{\text{OUT, A}} = (C_j * A_{\text{D, n}}) + (C_{\text{jsw}} * P_{\text{D, n}}) + (C_j * A_{\text{D, p}}) + (C_{\text{jsw}} * P_{\text{D, p}}) = 1.096 \text{ fF}$
- $C_{\text{LOAD}} \approx 3.216 \text{ fF}$
- $I_{\text{starve}} = -8.683 \text{ }\mu\text{A}$ (negative because going into NFET mirror)

Design Analysis – Calculating W/L for Starve FETs

- $I_{\text{starve}} = 8.683 \mu\text{A}$
- Want to bias the starve FETs to saturation region
- $V_{\text{OV}} = V_{\text{GS}} - V_{\text{TH}}$
- Want V_{OV} to put FETs in saturation region but also leave headroom. Pick semi-arbitrary value within those constraints of 0.2 V
- $I_{\text{starve}} = 0.5\mu\text{C}_{\text{ox}}(W/L)(V_{\text{OV}})^2$
- NMOS: $W/L = 1.64$
- PMOS: $W/L = 5.22$
- These ratios apply to the starving FETs as well as the mirror FETs. The starve and mirror transistors need to have the same ratio so that current is properly mirrored.
 - NMOS: 570n/350n
 - PMOS: 1.83u/350n

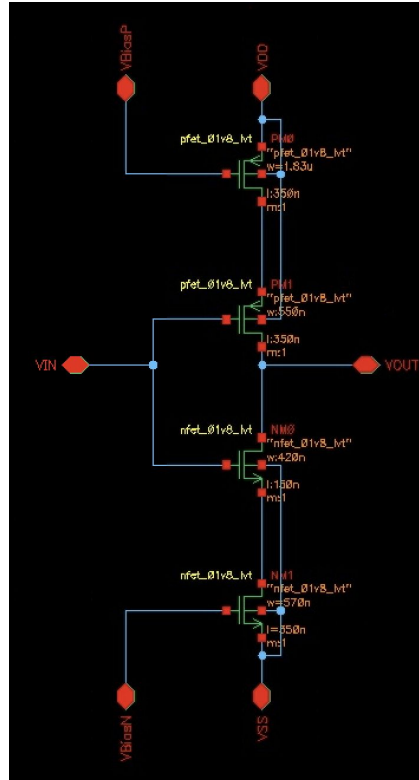
Design Analysis – Buffer

- 4 stages of inverters (must be even number as opposed to oscillator) with increasing sizes
- Match inverting transistors sizing for first stage, then multiply W for all NFETs by 4 per stage
- Maintain 1:1 ratio of $W_p:W_n$
 - Picked ratio by trial and error
 - Started with 2.5:1, duty cycle was too high, reduced until <60%

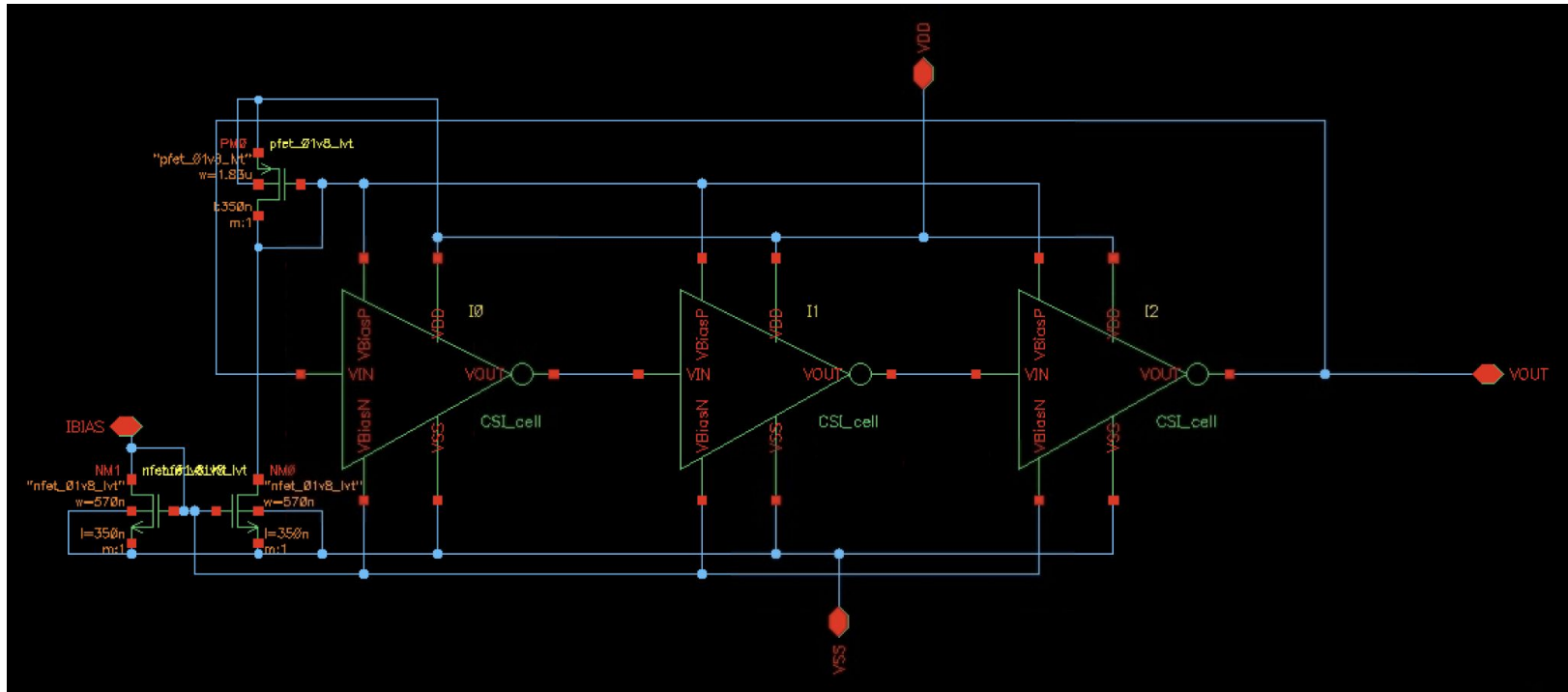
Design Analysis – Tuning

- After assembling and simulating, observed a frequency of 670 MHz using the numbers I calculated above. This is quite far from the expected 500 MHz, and it is due in part to my approximations in finding load capacitance. I disregarded parasitic capacitance from the transistors and used approximated values for C_{Ox} , C_j , C_{jsw}
- The fix is to simply tune I_{bias}
 - Achieved 502.0 MHz frequency with $I_{bias} = -6.413 \mu A$

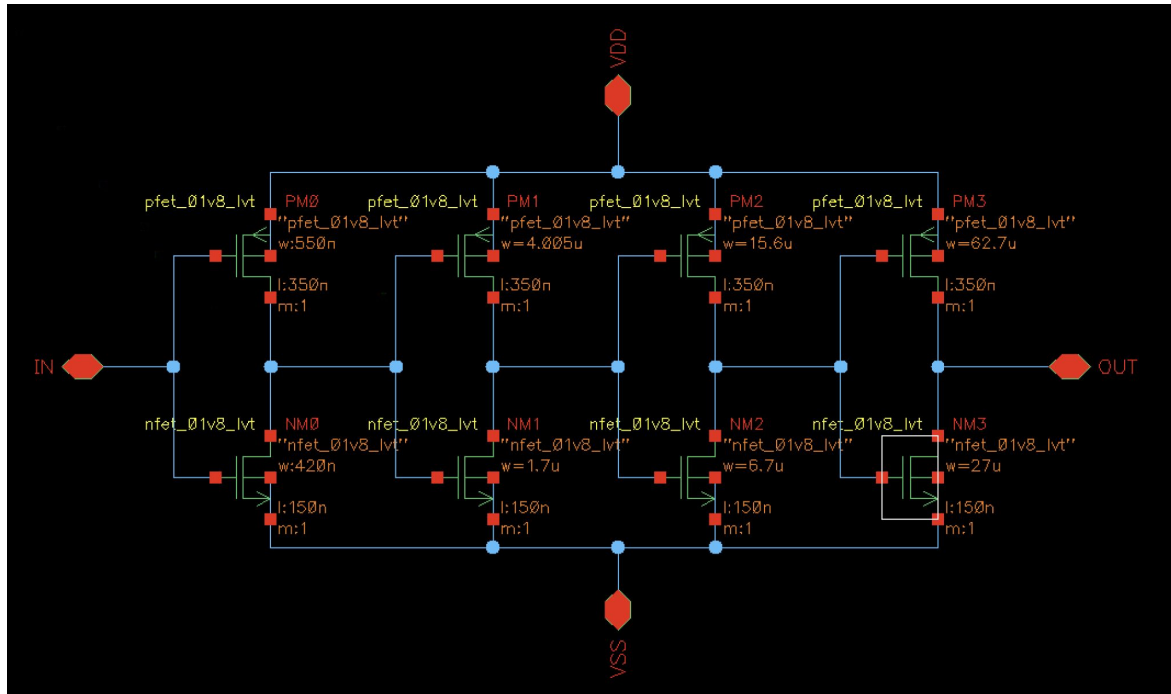
Design Schematics – Current Starved Inverter



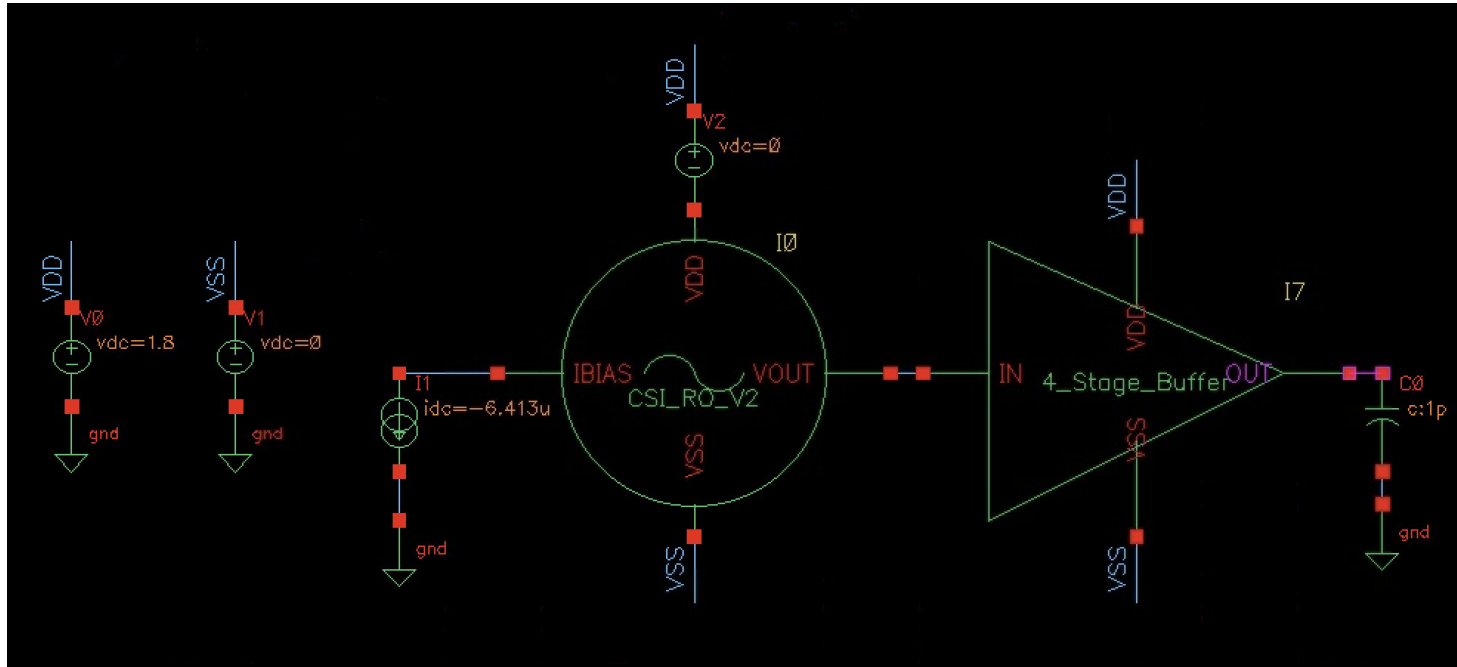
Design Schematics – Ring Oscillator



Design Schematics – Buffer



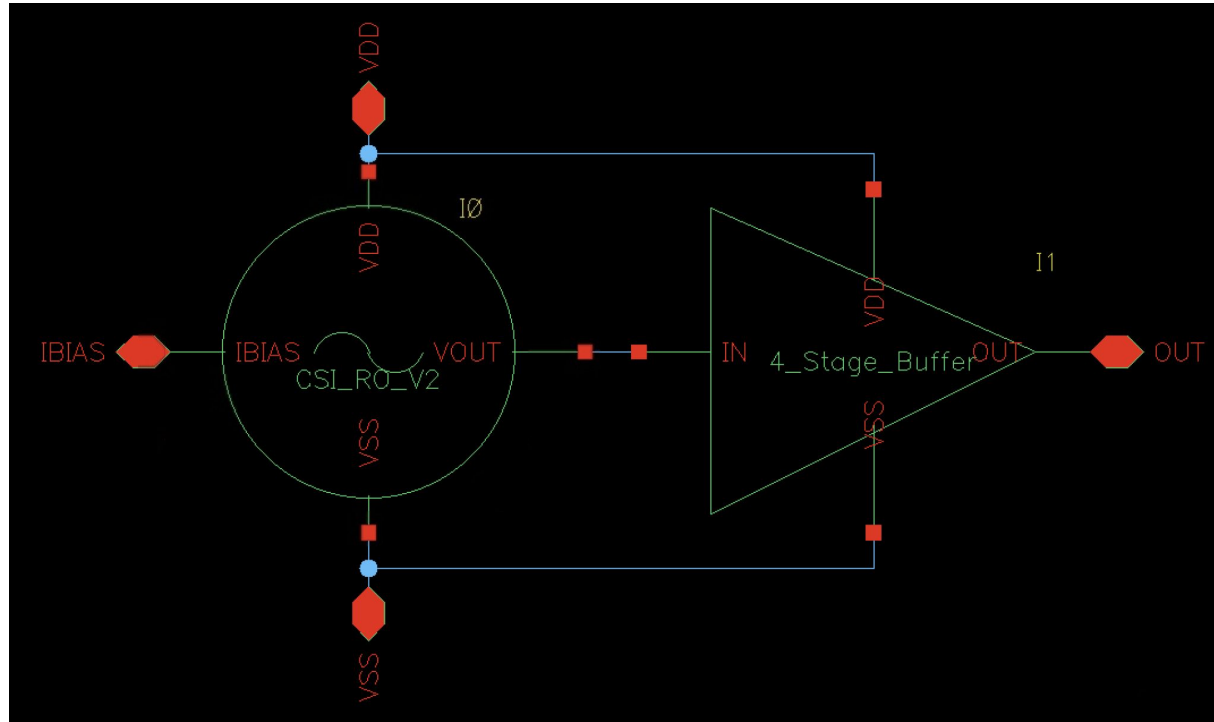
Design Schematics – Testbench



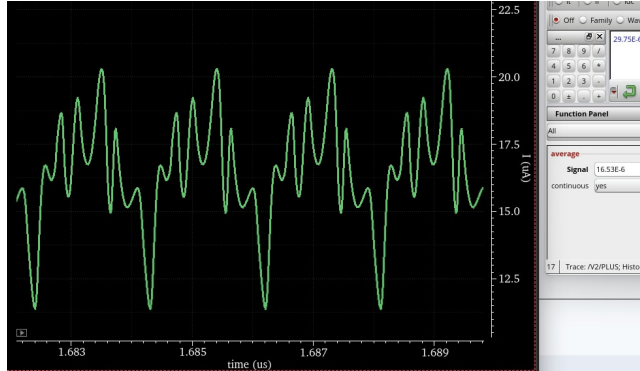
Specs met:

- VDD = 1.8 V
- GND = 0 V
- Cload = 1 pF

Design Schematics – Wrapper



Average Power Draw



0 C

- Multiplied average current by 1.8 V

Oscillator power @ 0 C = 29.75 μ W

@ 70 C = 31.55 μ W

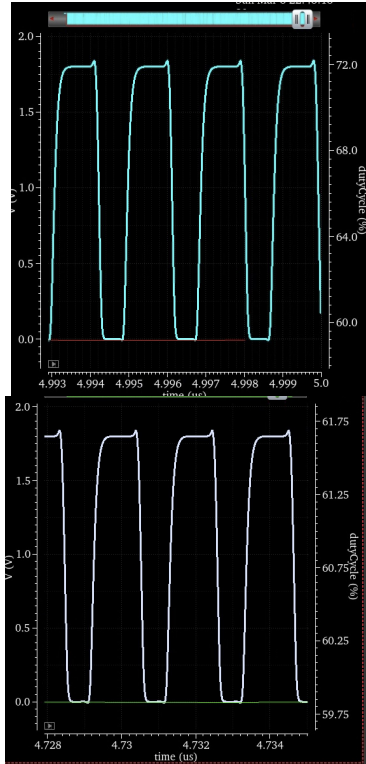


70 C

Specs met:

- Oscillator Power < 500 μ W

Duty Cycle and t_r / t_f

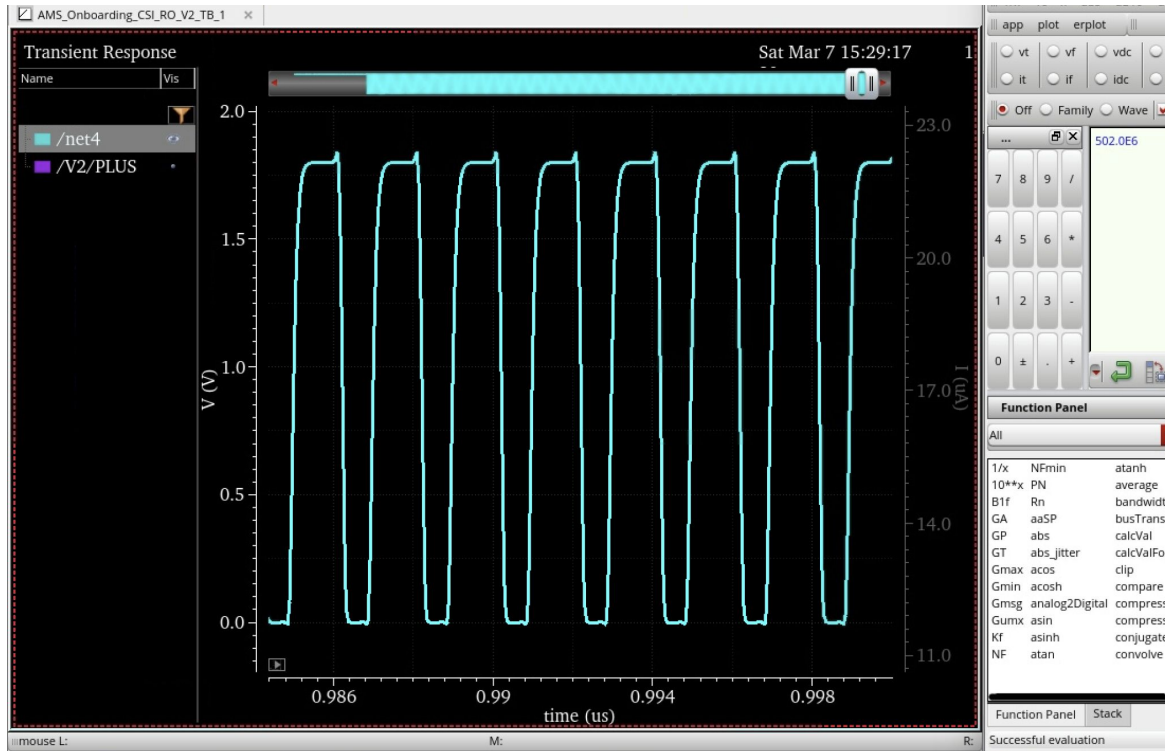


Specs met:

- Duty cycle:
 - @ 0 C = 59.1%
 - @ 70 C = 59.8 %
- 10% -> 90%:
 - @ 0 C = 215 ps
 - @ 70 C = 236.3 ps

Note: should have shrunk the buffer transistors more to more cleanly meet duty cycle spec, but I ran out of time

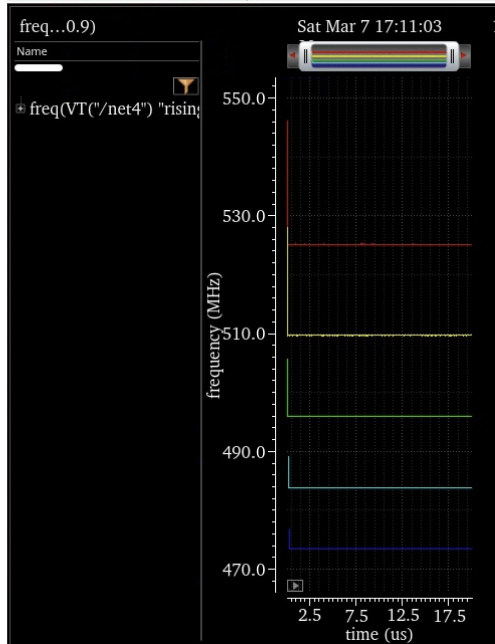
Oscillation at 27 C



Specs met:

- $f_{osc} = 502.0 \text{ MHz}$
- $V_{min} = 2.3 \text{ } \mu\text{V}$
- $V_{max} = 1.79 \text{ V}$

Frequency vs. Temperature

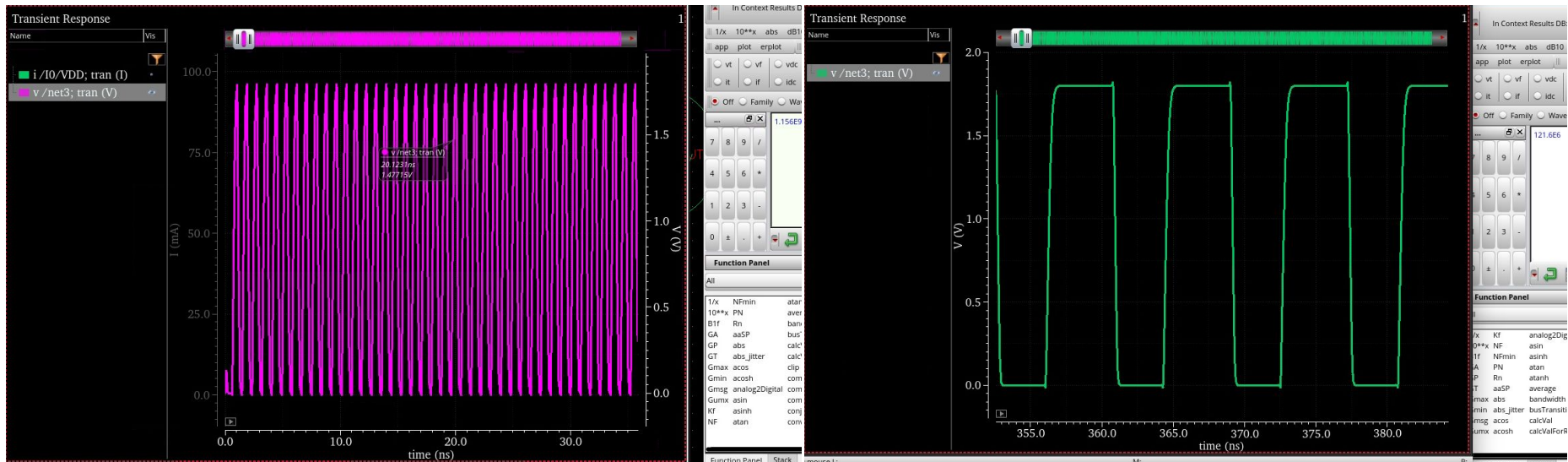


Specs met:

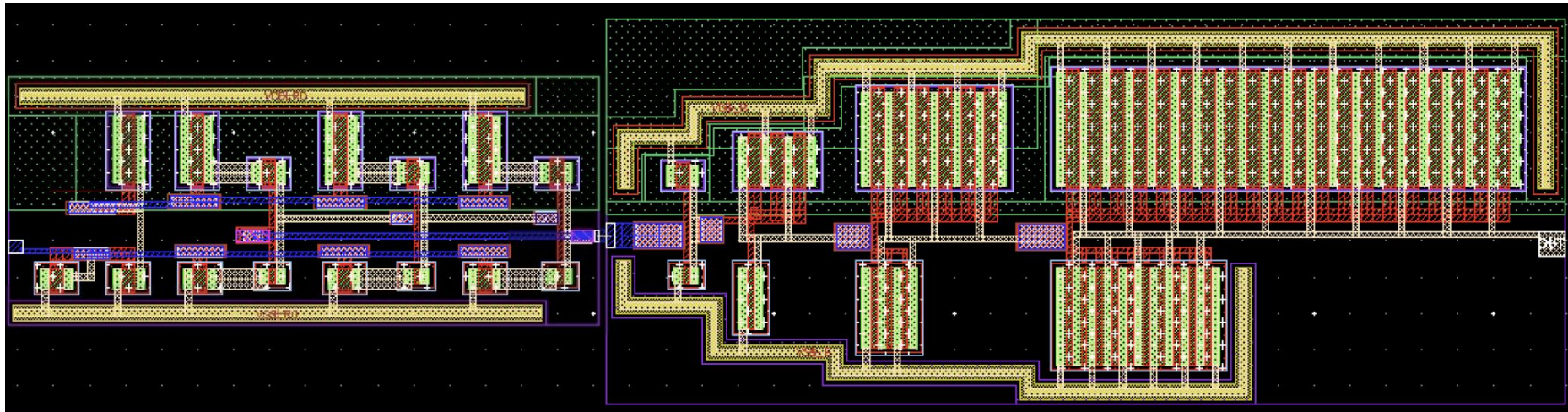
- Temperature
 - At 0 C f_{osc} is about 525 MHz and drops to around 472 MHz at 70 C

Note: chose to simulate 5 points to since the most important points are the start and end (0 C) and the middle ones are all between the extreme values. I didn't want to waste server resources or simulation time so I kept the number of points low.

ff and ss corners



Layout



Passed DRC

```
Worker resource usage summary: Total CPU[s]: 10 Run duration[s]: 13 Max peak memory[MB]: 688
Worker CPU utilization summary: 10 of 52 available CPU seconds (19.23%) with 4 average available CPUs
INFO: Generating summary in /nethome/nramam3/AMS-Onboarding/Cadence/Virtuoso/CSI_RO_V2_TOP.sum
INFO: All selected rules completed
      Total CPU Time           : 10(s)
      Total Real Time          : 13(s)
      Total Original Geometry  : 1406 (1718)
      Total DRC RuleChecks     : 428
      Total DRC Results        : 0 (0)
      ASCII report database is /nethome/nramam3/AMS-Onboarding/Cadence/Virtuoso/CSI_RO_V2_TOP.drc_errors.ascii
INFO: Checking in all SoftShare licenses.

Pegasus finished normally. 2026-03-08 21:17:19
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Passed LVS

